CIVORA

AI-Powered Citizen Grievance Redressal Platform

Tagline: "Making Public Grievance Redressal Smarter, Faster, and More Accountable."

ACADEMIC RESEARCH PROTOTYPE • CIVIC TECH PITCH
HUMAN-IN-THE-LOOP AI ARCHITECTURE
Submission Ready • Strictly 1 A4 Page Proposal

END-TO-END WORKFLOW: Citizen Intake (Text/Voice) → AI Relevance & NLP Classification → Biometric / OTP Verification → Department Triage & SLA Tracking → Admin GIS Analytics & Redressal

1. PROJECT BACKGROUND

CIVORA is an AI-powered digital grievance redressal and decision-support platform engineered to bridge the critical trust, speed, and communication gap between citizens and public authorities. It establishes a centralized, transparent system for submitting, categorizing, routing, tracking, and managing municipal and civic grievances.

By harmonizing **NLP-based classification, voice-assisted intake, facial biometrics, and administrative analytics**, CIVORA transforms conventional bureaucratic friction into responsive, data-driven governance while rigorously upholding human administrative oversight.

2. CORE PROBLEMS ADDRESSED

- **Delayed Processing:** Overwhelming complaint backlogs and protracted manual review cycles.
- **Manual Routing Errors:** High rate of inter-departmental misdirection and administrative ping-pong.
- **Opacity & Distrust:** Complete absence of real-time grievance tracking and verifiable audit logs.
- **Hidden Recurring Clusters:** Inability to correlate repetitive civic failures across municipal wards.
- **Digital & Literacy Barriers:** Intricate text-only web forms exclude illiterate, elderly, and rural citizens.
- **Administrative Blindspots:** Lack of centralized telemetry, SLA tracking, and GIS spatial intelligence.

3. PROPOSED SOLUTION & IMPLEMENTED CAPABILITIES

Citizen Grievance Management
- Secure citizen registration, JWT sessions, and RBAC security.
- Rich complaint filing: text descriptions + image evidence via Cloudinary.
- Unique alphanumeric tracking reference IDs with verifiable audit trails.
- Comprehensive real-time status, timeline, and resolution history tracking.

AI-Powered Classification & Triage
- Dual NLP engine: TF-IDF + Logistic Regression & Hugging Face pipeline.
- Automated relevance validation gatekeeper rejecting spam or non-civic text.
- Automated department routing and severity prioritization scoring.
- Human-in-the-loop design: AI advises; administrative personnel approve.

Voice-Assisted Conversational Intake
- Native speech-to-text audio transcription powered by GNANI Prisma STT.
- Structured entity extraction parsing location, incident type, and urgency.
- Dynamic contextual follow-up questions prompting for missing complaint details.
- Voice feedback via GNANI Timbre TTS, democratizing access for non-literate citizens.

Secure Facial Biometrics & Liveness Verification
- On-device facial detection & recognition with InsightFace (ONNX Runtime engine).
- High-efficiency model pipeline using SCRFD detector and ArcFace/MobileFaceNet models.
- Active liveness verification challenges to prevent photo/screen presentation attacks.
- Biometric privacy: 512-d embeddings encrypted with Fernet before database storage.

Administrative Intelligence Console & GIS Analytics
- Centralized command portal monitoring city-wide grievances across departments.
- Real-time departmental complaint distribution and workload balancing.
- Interactive charts and analytical visualizations rendered via Recharts.
- Geographic spatial cluster and hotspot mapping powered by Leaflet GIS.
- SLA compliance monitoring, escalation triggers, and resolution verification.

4. TECHNOLOGY STACK (PRODUCTION-PROVEN & INTEGRATED)

Frontend: Next.js 16 (App Router), React 19, JavaScript, Tailwind CSS	**Backend:** Python 3.12+, FastAPI, Uvicorn, Pydantic, Requests	**Database:** MongoDB Atlas (Cloud NoSQL, Geospatial & Text Indices)
AI / NLP: TF-IDF, Logistic Regression, Groq (Qwen-27B), Hugging Face	**Biometrics:** InsightFace, ONNX Runtime (SCRFD + MobileFaceNet)	**Voice AI:** GNANI Prisma STT (Speech-to-Text) & GNANI Timbre TTS
Auth & Sec: JWT, Role-Based Access (RBAC), Phone OTP, Fernet Encryption	**GIS & Viz:** Recharts (Interactive Dashboards) & Leaflet (Geospatial GIS)	**Deployment:** Vercel (Frontend CDN), Render (FastAPI Cloud), Cloudinary

5. INNOVATION & DIFFERENTIATION

- **Relevance-Validated Triage:** Automated screening filters spam & irrelevant submissions before routing.
- **Conversational Voice Intake:** Dynamic contextual questioning captures missing civic parameters hands-free.
- **Anti-Spoof Biometrics:** On-device ONNX face recognition with active liveness prevents identity fraud.
- **Human-in-the-Loop AI:** Automated systems assist prioritization without usurping human administrative authority.
- **Unified Full Lifecycle:** Merges citizen tracking, multi-department routing, and administrative oversight.

6. EXPECTED SOCIETAL IMPACT

- **Eliminates Processing Delays:** Instant categorization reduces manual sorting and routing backlogs.
- **Accurate Department Routing:** Minimizes misdirected complaints and inter-departmental runarounds.
- **Universal Accessibility:** Multimodal voice intake empowers semi-literate, elderly, and rural citizens.
- **Radical Public Accountability:** Unique IDs & tamper-evident audit history rebuild civic trust.
- **Proactive Civic Maintenance:** GIS heatmaps enable data-driven resource allocation over reactive fixes.

7. FUTURE SCOPE (ROADMAP)

- **Multilingual Voice Expansion:** Native support for 12+ regional Indian languages and local dialects.
- **Predictive Civic Analytics:** Machine learning to anticipate recurring municipal faults & SLA risks.
- **Government API Gateway:** Direct integration with CPGRAMS, state portals, and Smart City command centers.
- **Enterprise Scalability:** Distributed microservice expansion for state- and nation-wide deployment.
- **Resolution Quality AI:** Automated citizen satisfaction and resolution-adequacy verification.